BIOCHEMISTRY CUME
October 8, 2004

You have isolated one pg (10^{-12} gram) of a circular double-stranded DNA that contains 5555 base pairs. The chemically pure sample of DNA is dissolved in ten microliters of water, appropriately buffered at low ionic strength.

1. (15 points) To the nearest 1 million molecules, how many DNA molecules do you have?

2. (15 points) You decide to examine your sample using agarose gel electrophoresis. Even though the sample is chemically pure, you observe at least two DNA bands after staining the gel with ethidium bromide and stimulating the ethidium fluorescence with UV light. Give the most plausible explanation as to <u>why</u> more than one band of DNA is observed.

3. (15 points) A colleague shows you a small tube and suggests that you add one microliter of the solution from the tube to your DNA sample. You follow the suggestion, rerun the agarose gel electrophoresis, and then observe only one band on the gel. What was in your colleague's tube?

4. (55 points) You are interested in a particular gene consisting of a known sequence of 1113 base pairs that is present within your isolated DNA sample. For future experiments, you wish to synthesize one mg (10^{-3} gram) of the DNA corresponding to this gene. How will you do it? Be specific about the methods, procedures, equipment, chemicals, amounts of particular chemicals, and any or all other supplies that you will need. The grader of this question should be able to take your materials into a laboratory equipped according to your specifications, follow your procedures and emerge with one mg of the desired DNA.

Supplemental information: For the dinucleotide-phosphate, GpCp, the approximate molecular formula is: $C_{19}H_{18}N_8O_{14}P_2$.